

$$\begin{array}{l} \nabla\times\\ \mathbf{E}=\frac{\partial\mathbf{B}}{\partial t},\nabla\times\\ \mathbf{H}=\frac{\partial\mathbf{D}}{\partial t}+\\ \mathbf{J},\mathbf{B}=\\ \mu\mathbf{H},\mathbf{D}=\\ \varepsilon\mathbf{E}\\ ?\\ \Delta t\leq\frac{1}{c\sqrt{(1/\Delta x)^2+(1/\Delta y)^2+(1/\Delta z)^2}}\\ \Delta t\\ \Delta x\Delta y\Delta z\\ ?\\ ??\\ ??\\ \overline{2.40}\\ WS_2\\ \overline{2.40}\\ 1/resolution\\ \overline{3.31}\\ \overline{1}\\ t_{design}=\\ 0.257\,\mu m\\ \overline{\lambda}=0.593\,\mu m\\ 2\pi\\ WS_2\\ S^x=\\ S^y=\\ W_{design}+\\ 2\,d_{pml}\\ W_{design}=\\ 4.5\,\mu m\\ d_{pml}=\\ 0.3\,\mu m\\ 50\,gridpoints/\mu m\\ N_x\times\\ N_y=\\ 226\times\\ 226\approx\\ 5.1\times\\ 10^4\in\\ \rho_j\in\\ [0,1]\\ \overline{\lambda}=0.593\,\mu m\\ E_g\approx\\ 2.26\,\mathrm{eV}\\ _structure_{waist}.png\\ _structure.png\\ \overline{2.40}\\ \overline{1}\\ t_{design}=\\ 0.257\,\mu\\ WS_2\\ WS_2\\ ,\\ {}_0\big[\hat{x}\,e^{-i\omega_0t}+i\,\hat{y}\,e^{-i\omega_0t}\big]\,f(t)\\ f(t)\\ \hat{x}\\ \hat{y}\\ 90^\circ\\ \sigma_+\\ WS_2\\ ??\\ ?\\ \dot{\varepsilon}_0\,\omega\,\frac{1}{2}\overline{Im(\mathbf{E}^*\cdot\mathbf{H})}\\ \varepsilon_0=\\ \mu_0=\\ \zeta=\\ 1\omega=\\ 2\pi f\\ \pi f\,Im(\mathbf{E}^*\cdot\mathbf{H})\\ C>\\ 0\\ C<\\ 0\\ \overline{C_{CPL,vac}}\\ C_{CPL,vac}\\ ?\\ \Rightarrow\\ \Rightarrow\\ \Rightarrow\\ ??\\ \rho(\mathbf{x})\in\\ [0,1]\\ ?\end{array}$$